

TECHNOLOGY TRENDS & DEVELOPER DEMOGRAPHICS ANALYSIS

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OUTLINE



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EXECUTIVE SUMMARY



This analysis examines developer technology trends using Stack Overflow survey data. Results show continued dominance of Python, SQL, and JavaScript, rising cloud and open-source database adoption, and a positive relationship between experience, compensation, and job satisfaction.



INTRODUCTION



- Purpose: Analyze current and future technology trends.
- Audience: Data professionals and technology leaders.
- Value: Supports skill and hiring decisions.

METHODOLOGY



- **Data Sources:**

- Stack Overflow Developer Survey
- Web-scraped HTML dataset
- Public REST API

- **Tools:**

- SQL, Python, IBM Cognos Analytics

- **Steps:**

- Cleaning and preprocessing
- Normalization and transformation
- Analysis and visualization

RESULTS

Languages: JavaScript, SQL, and Python lead adoption

Databases: PostgreSQL and MySQL dominate usage

Platforms: AWS, Azure, and Google Cloud are most used

Demographics: Majority of developers are primarily aged 25–44

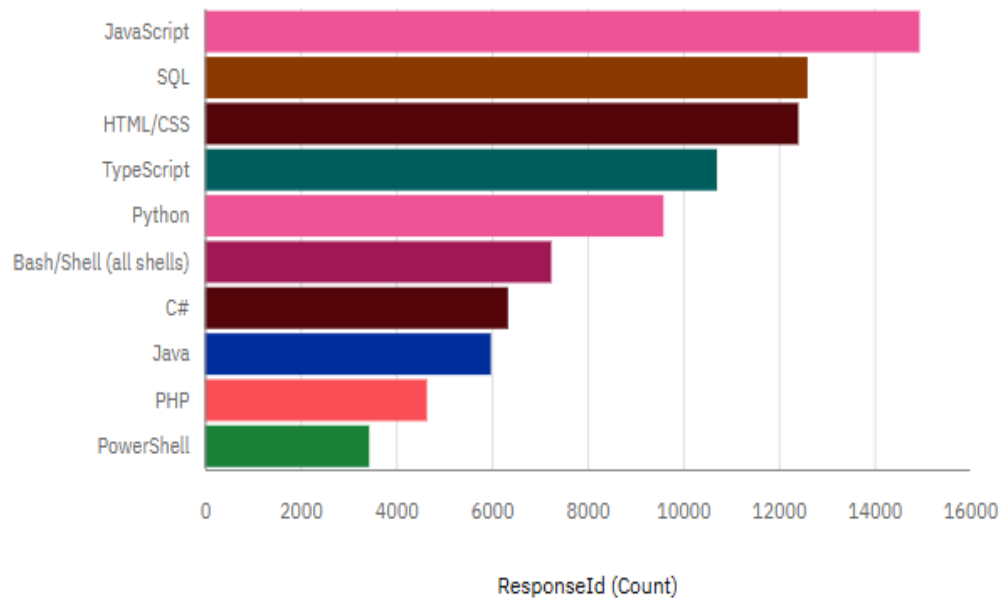
PROGRAMMING LANGUAGE TRENDS

Current Year

Top 10 Programming Languages Developers Worked with



LanguageHaveWorkedWith

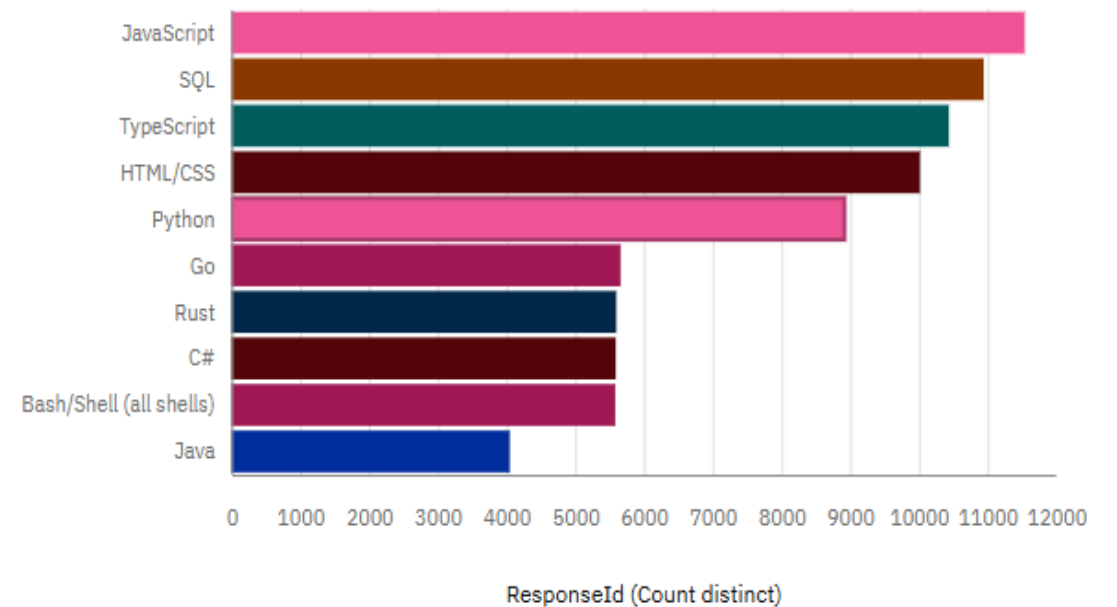


Next Year

Top 10 Programming Languages Developers Want to Work With



LanguageWantToWorkWith



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JS, SQL, Python lead
- TypeScript rising
- Go, Rust emerging

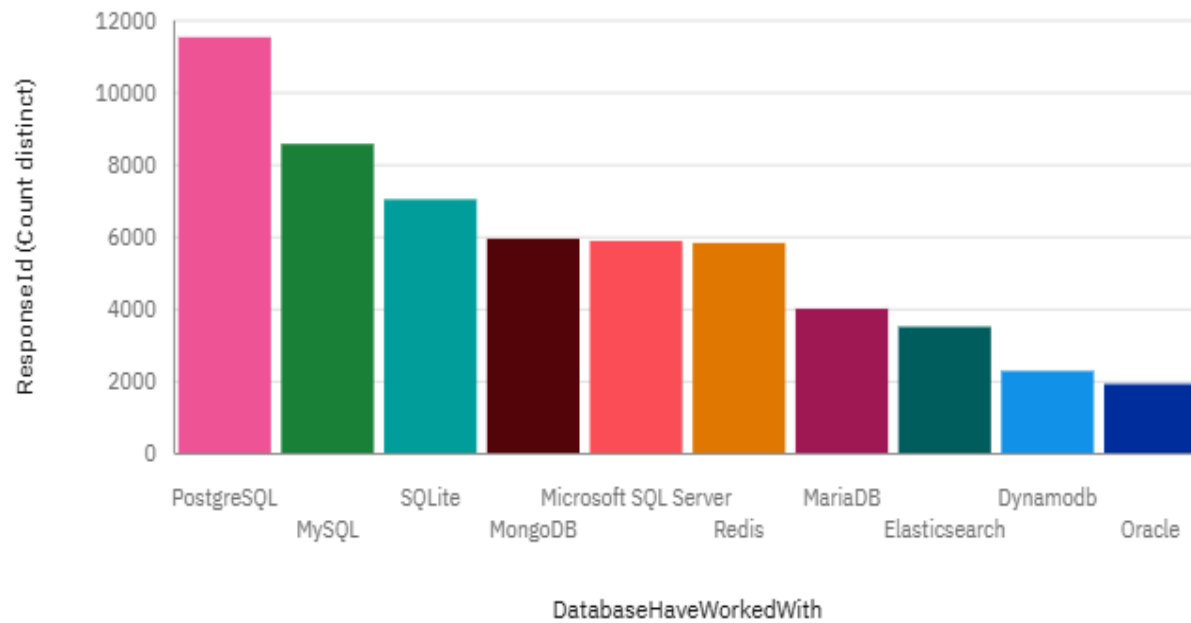
Implications

- Strong demand for core skills
- Shift toward modern stacks
- Need for ongoing learning

DATABASE TRENDS

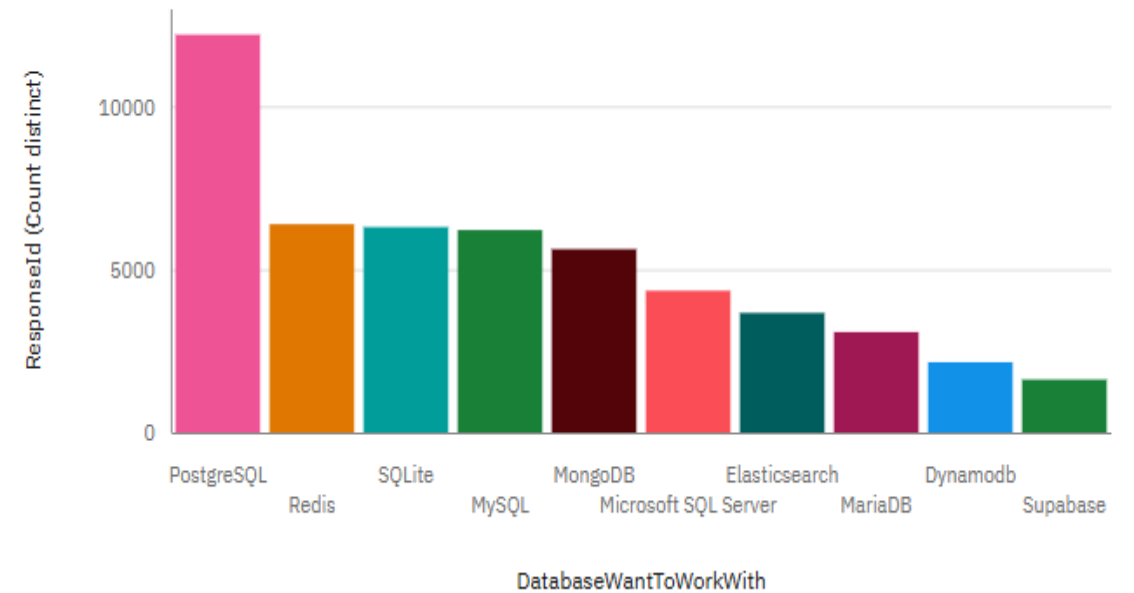
Current Year

Top 10 Databases Developers Worked With



Next Year

Top 10 Databases Developers Want to Work With



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL, MySQL lead
- MongoDB popular
- Redis growing

Implications

- SQL skills remain essential
- NoSQL adoption increasing
- Focus on scalability & speed



DASHBOARD



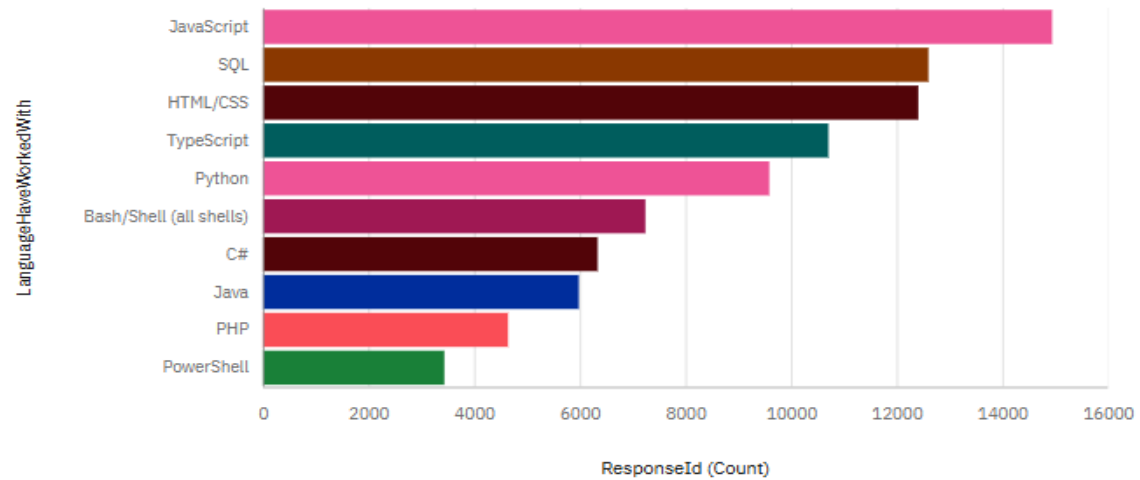
- Current Technology Usage
- Future Technology Trends
- Demographics

https://github.com/mythiliskiran/in-demand-developer-skills-analysis/blob/b8bbebda9ac9856b8fbe468095664da2bd9723f5/building_a_dashboard/Stack%20Overflow%20Survey%20dashboard.pdf

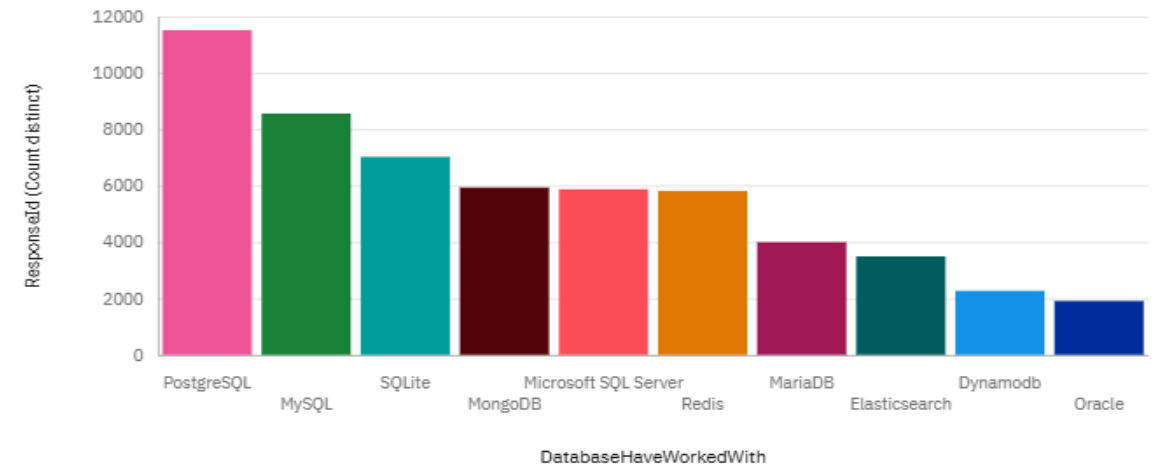


CURRENT TECHNOLOGY USAGE

Top 10 Programming Languages Developers Worked with



Top 10 Databases Developers Worked With



Top 10 Platforms Worked With

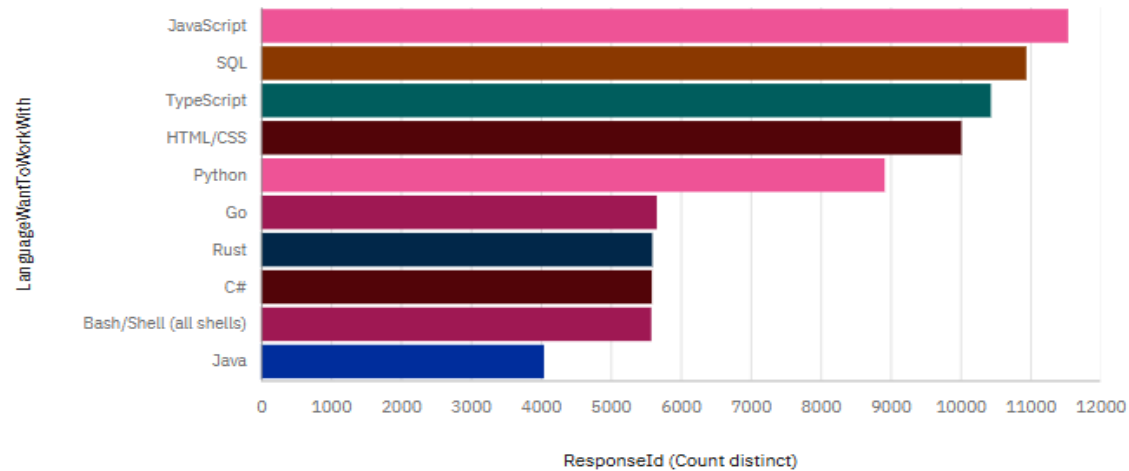


Top 10 Web Frameworks Worked With



FUTURE TECHNOLOGY TRENDS

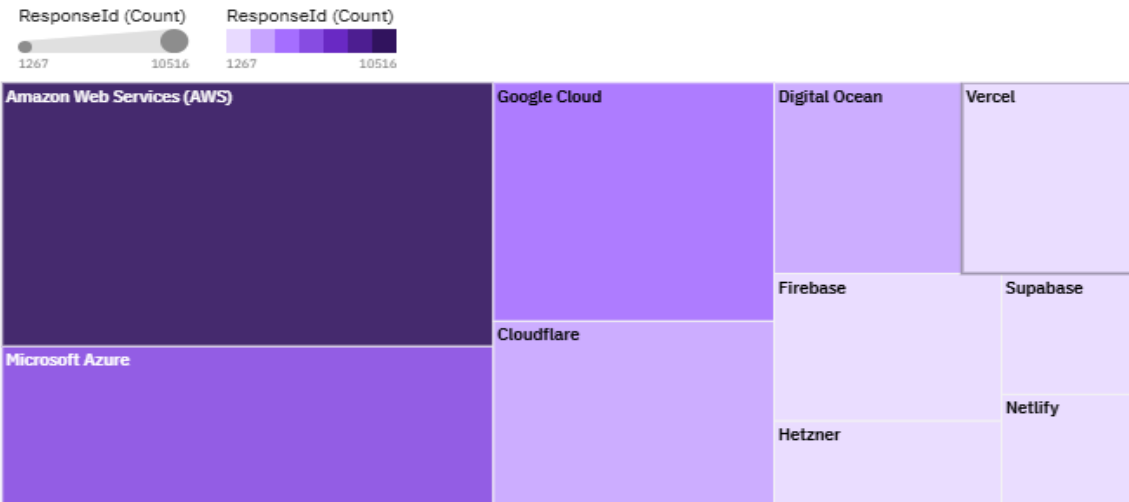
Top 10 Programming Languages Developers Want to Work With



Top 10 Databases Developers Want to Work With



Top 10 Platforms Developers Want to Work With

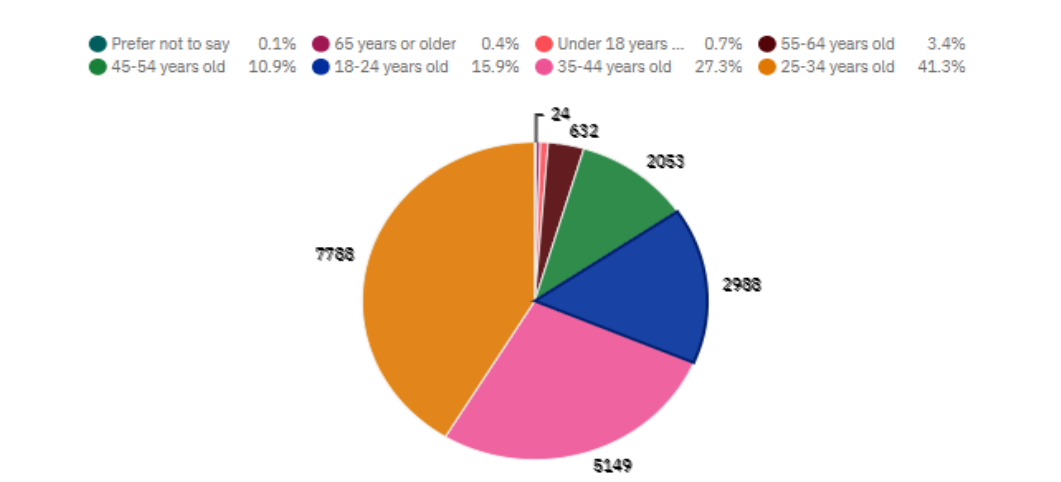


Top 10 Web Frameworks Developers Want to Work With

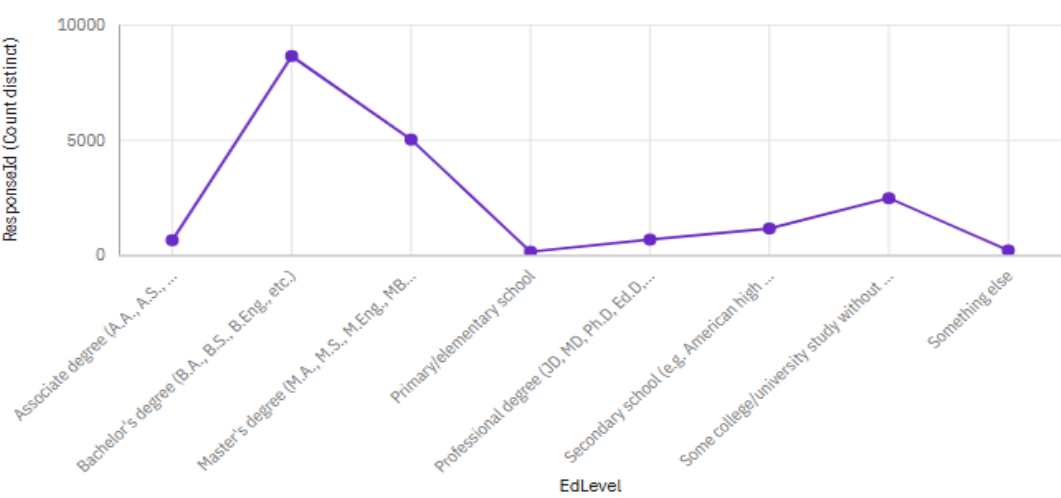


DEMOGRAPHICS

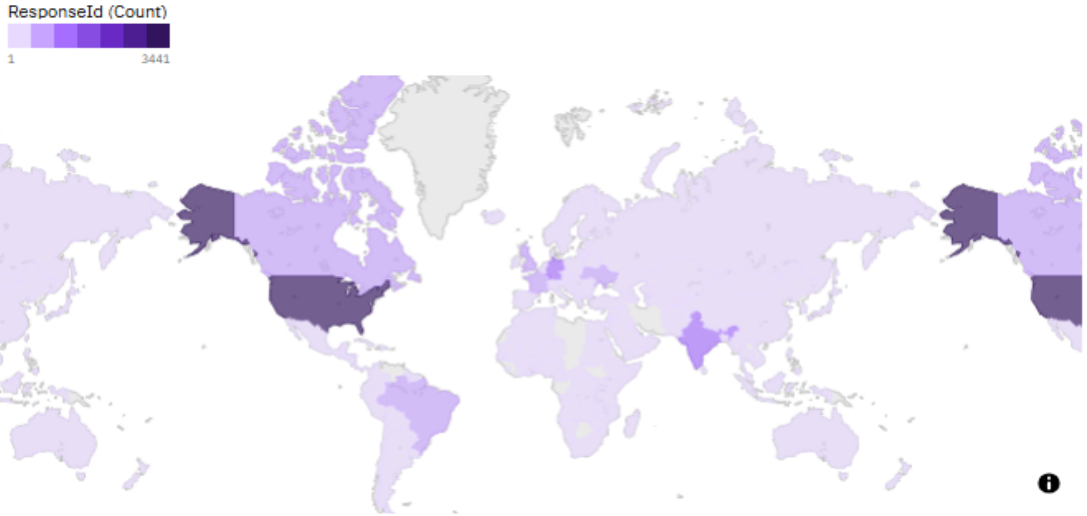
Respondent Distribution by Age



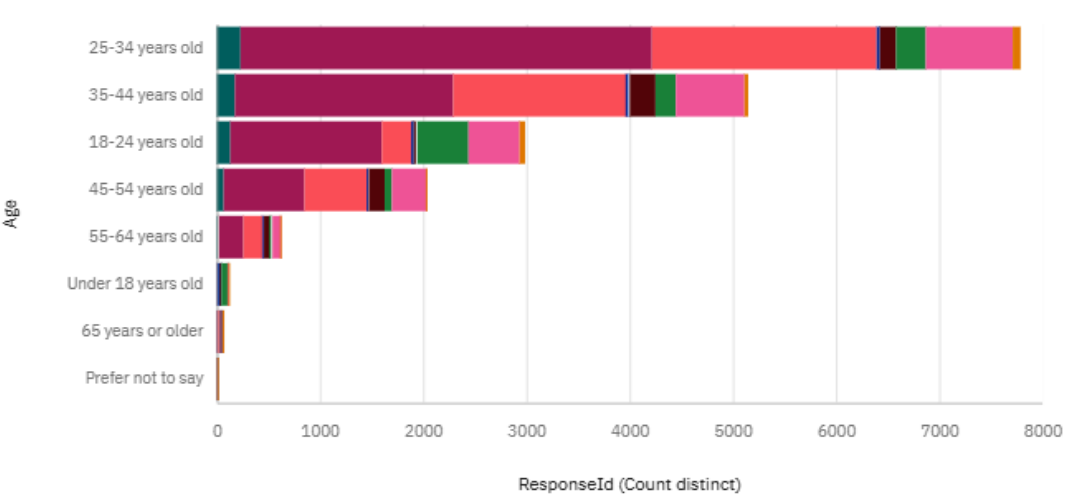
Respondent Distribution by Formal Education Level



Respondent Count by Country



Respondent Count by Age and Education Level



DISCUSSION



- Core technologies remain dominant across roles
- Cloud adoption reflects industry scalability needs
- Emerging tools show gradual, not disruptive, growth
- Skills demand driven by both stability and innovation

OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript, SQL, Python dominate usage
- Cloud platforms widely adopted
- Relational + NoSQL databases both important

Implications

- Core technical skills remain essential
- Cloud and data skills drive demand
- Continuous upskilling is critical

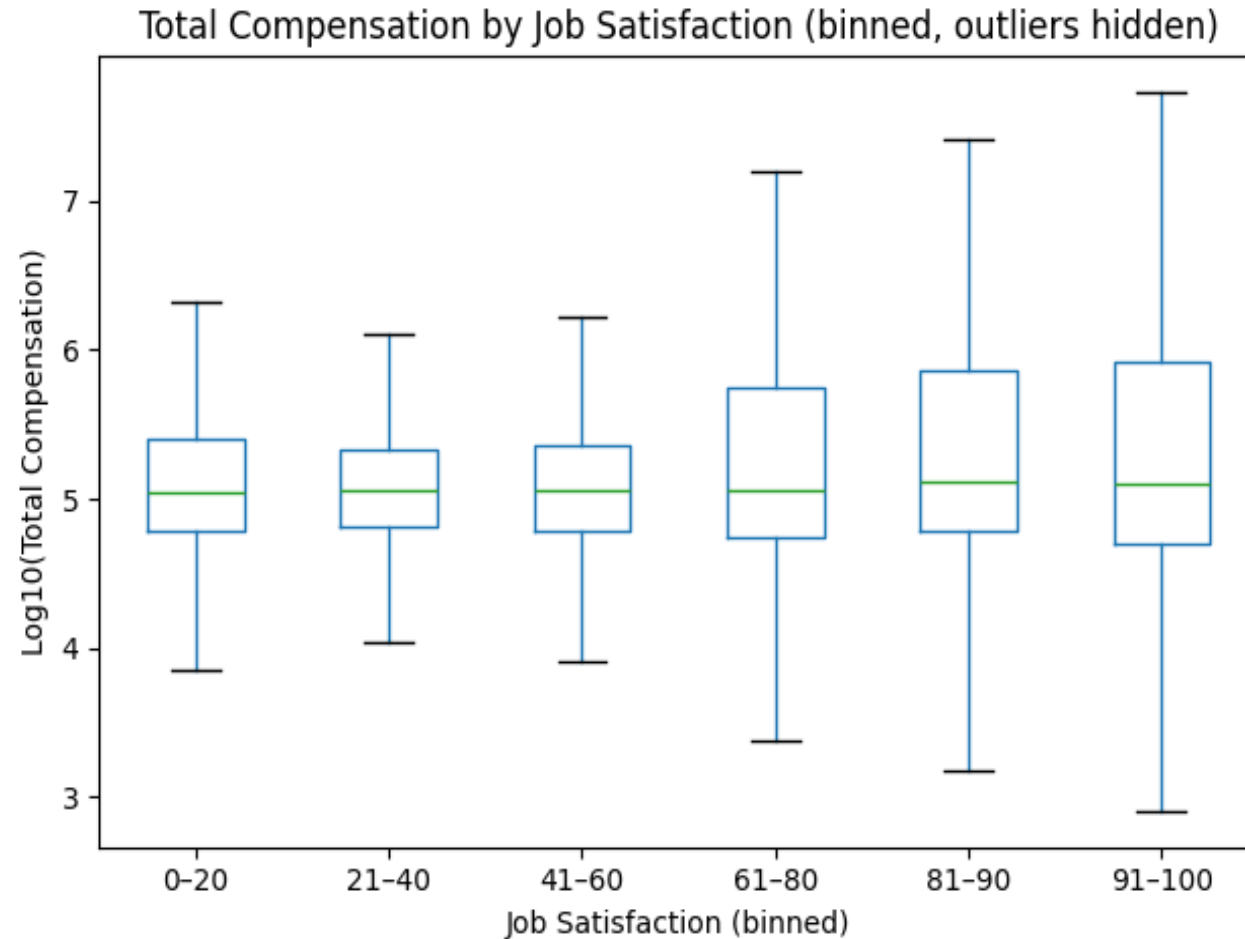
CONCLUSION



- Core programming and database technologies remain essential
- Cloud platforms play a central role in modern development
- Emerging tools signal evolving industry needs
- Continuous learning is key to staying relevant

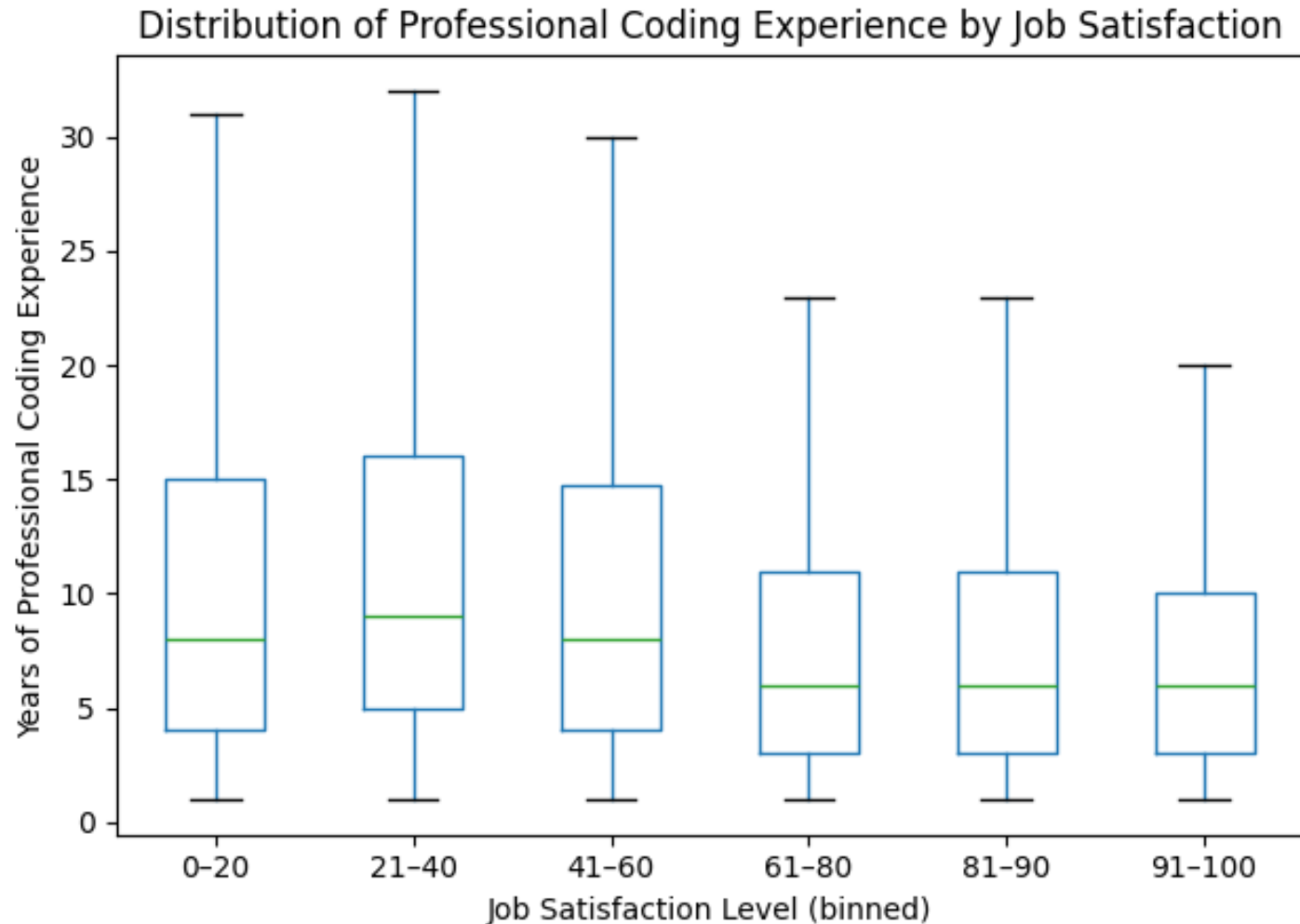


APPENDIX A – COMPENSATION VS JOB SATISFACTION



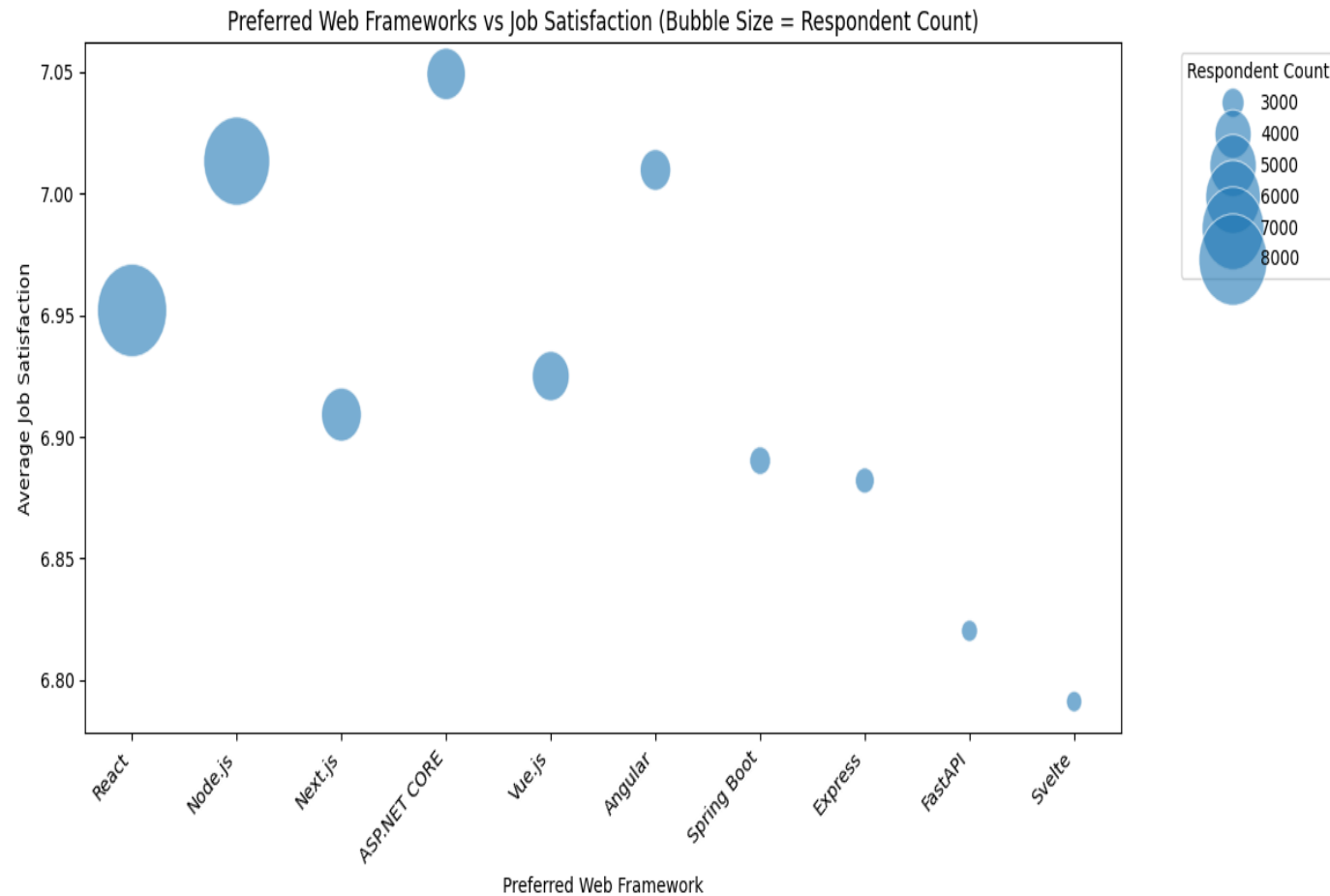
Median compensation shows a slight upward trend with higher job satisfaction, while salary variability increases at higher satisfaction levels.

APPENDIX B – EXPERIENCE VS JOB SATISFACTION



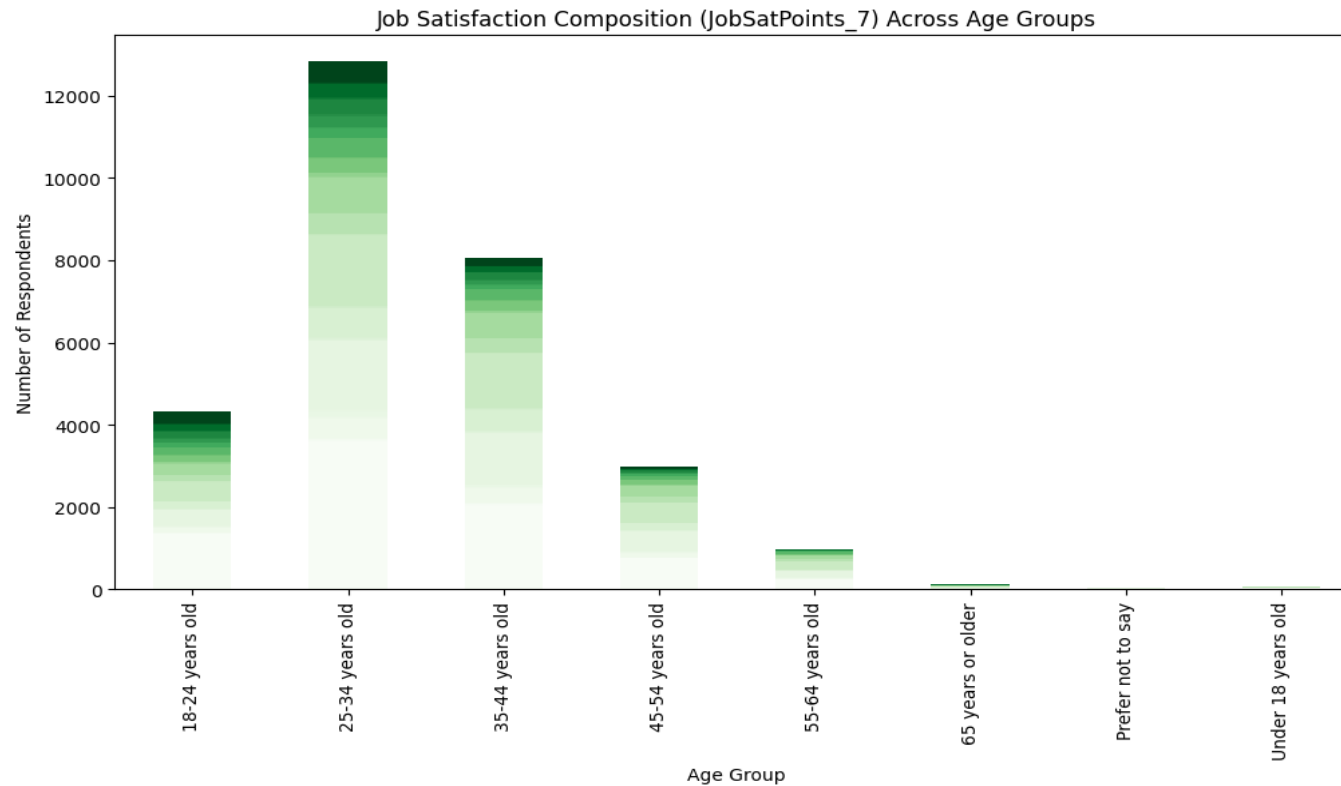
Higher job satisfaction groups tend to have slightly fewer years of professional coding experience, with wider experience ranges at lower satisfaction levels.

APPENDIX C – PREFERRED WEB FRAMEWORKS JOB SATISFACTION



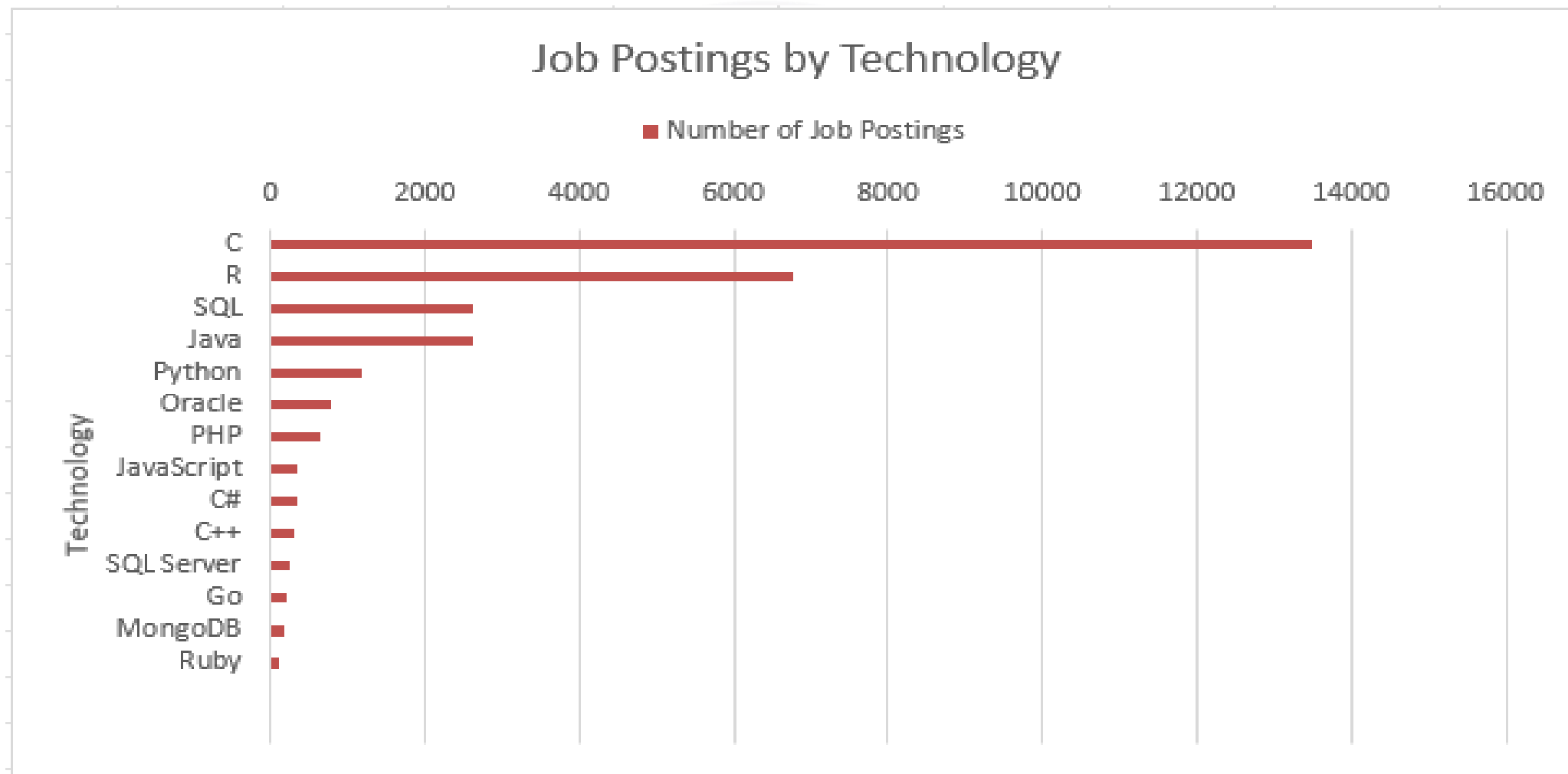
The bubble plot shows that although some web frameworks are more popular than others, average job satisfaction remains relatively similar across framework preferences.

APPENDIX D – JOB SATISFACTION ACROSS AGE GROUPS



Job satisfaction is highest in volume among mid-career respondents, with comparable satisfaction distributions across age groups.

APPENDIX E – JOB POSTINGS



APPENDIX F – POPULAR LANGUAGES

